US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12400707
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Dhori; Kedar Janardan

Device and method for reading data from memory cells

Abstract

An integrated circuit includes a memory array. The memory array includes a plurality of bitlines. The bitlines are each coupled to a respective local I/O circuit. All of the local I/O circuits are coupled to a global I/O circuit. Each local I/O circuit includes a first sensing stage for reading data from the memory cell. The global I/O circuit includes a second sensing stage for reading data from the memory cell.

Inventors: **Dhori; Kedar Janardan (Ghaziabad, IN)**

Applicant: **STMicroelectronics International N.V.** (Geneva, CH)

Family ID: **1000008782830**

Assignee: **STMicroelectronics International N.V. (Geneva, CH)**

Appl. No.: **17/494683**

Filed: **October 05, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20220122657 A1	Apr. 21, 2022

Related U.S. Application Data

us-provisional-application US 63092305 20201015

Publication Classification

Int. Cl.: **G11C11/419** (20060101); **G11C11/412** (20060101); G11C11/418 (20060101)

U.S. Cl.:

CPC **G11C11/419** (20130101); **G11C11/412** (20130101); G11C11/418 (20130101)

Field of Classification Search

CPC: G11C (11/418); G11C (11/419); G11C (11/412)

USPC: 365/156

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5590087	12/1995	Chung	365/49.1	G11C 8/16
6944066	12/2004	Jing	N/A	N/A
8625373	12/2013	Schreiber	N/A	N/A
8792267	12/2013	Chandwani et al.	N/A	N/A
9165619	12/2014	Song	N/A	G11C 7/12
10446201	12/2018	Goel et al.	N/A	N/A
11094355	12/2020	Simon	N/A	G11C 7/065
2007/0133313	12/2006	Kim	365/207	G11C 7/1066
2008/0316840	12/2007	Kim	365/208	G11C 7/1096
2012/0317374	12/2011	Chen	711/E12.001	G11C 7/18
2020/0405254	12/2019	Kunieda	N/A	A61B 6/542

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103594107	12/2013	CN	N/A
105719689	12/2015	CN	N/A
110660430	12/2019	CN	N/A

Primary Examiner: Cho; Sung Il

Attorney, Agent or Firm: Seed Intellectual Property Group LLP

Background/Summary

BACKGROUND

Technical Field

(1) The present disclosure relates to the field of integrated circuit memory cells. The present disclosure relates more particularly to read operations of memory cells.

Description of the Related Art

(2) Integrated circuits often include arrays of memory cells. For many integrated circuits, it is beneficial for memory read and write operations to be performed at higher speed as this affects system-on-chip (SoC) performance. However, factors such as parasitic capacitance in bitlines can limit the speed of memory read and write operations. Efforts to reduce the negative effects of parasitic capacitance typically lead to other drawbacks.

BRIEF SUMMARY

(3) In one or more embodiments, the present disclosure provides an integrated circuit including a

multi-port memory cell configured to store a data value. The integrated circuit includes a plurality of bitlines each coupled to the multiport memory cell. The integrated circuit includes a plurality of local I/O circuits each coupled to a respective bitline and including a first sensing stage configured to sense the data value from the bitline when the bitline is selected for a read operation. The integrated circuit includes a global I/O circuit coupled to each of the local I/O circuits. The global I/O circuit includes a global bitline that can be selectively coupled to receive the data value from the local I/O circuit coupled to the bitline selected for the read operation. The global I/O circuit includes a second sensing stage including an evaluation circuit configured to sense the data value on the global bitline.

(4) In one or more embodiments, an integrated circuit includes a memory array having a plurality of banks of memory cells and a plurality of local I/O circuits each coupled to a respective bank of memory cells and including a first sensing stage configured to sense a data value from the respective bank of memory cells when the bank of memory cells is selected for a read operation. The integrated circuit includes a global I/O circuit coupled to each of the local I/O circuits. The global I/O circuit includes a global bitline that can be selectively coupled to receive the data value from the local I/O circuit coupled to the bank of memory cells selected for the read operation. The global I/O circuit includes a second sensing stage including an evaluation circuit configured to sense the data value on the global bitline.

(5) In one or more embodiments, a method includes storing a data value in a memory cell, selecting the memory cell for a read operation, and outputting the data value to a bitline coupled to the memory cell. The method includes sensing the data value from the bitline with a local I/O circuit coupled to the bitline, providing the data value from the local I/O circuit to a global I/O circuit by selectively enabling the local I/O circuit, and sensing the data value with the global I/O circuit and outputting the data value from the global I/O circuit.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. **1** is a block diagram of an integrated circuit including a memory array, according to one or more embodiments.

(2) FIG. **2** is a block diagram of a memory array including multiport memory cells, according to one or more embodiments.

(3) FIG. **3** is a block diagram of a memory array in the banking configurations, according to one or more embodiments.

(4) FIG. **4** is a schematic diagram of memory reading circuitry of an integrated circuit, according to one or more embodiments.

(5) FIG. **5** is a timing diagram of signals related to a memory read operation, according to one or more embodiments.

(6) FIG. **6** is a schematic diagram of memory reading circuitry of an integrated circuit, according to one or more embodiments.

(7) FIG. **7** is a flow diagram of a method for reading data from a memory cell from an integrated circuit, according to one embodiment.

DETAILED DESCRIPTION

(8) FIG. **1** is a block diagram of an integrated circuit **100**, according to one or more embodiments. The integrated circuit **100** includes a memory array **102**. The memory array **102** can include a plurality of memory cells **104**. For simplicity, FIG. **1** illustrates only a single memory cell **104**. The integrated circuit **100** further includes local I/O (LIO) circuits **106** and—global I/O (GIO) circuits **108**. As will be set forth in more detail below, the LIO circuits **106** and the GIO circuits **108** are configured to enable rapid read operations of the memory cells **104** while reducing the area used in

comparison to traditional I/O circuits.
(9) In some embodiments, the memory cell **104** is a multiport memory cell. A multiport memory cell is a memory cell that can output data to multiple ports. In one example, the integrated circuit **100** includes a multicore processor (not shown). Each core of the processor may read data from a same memory cell. Accordingly, the memory cell may be connected to a separate bitline for each core. This is one example of a multiport memory cell.
(10) The integrated circuit includes a group **120** of four bitlines, BL**1**-BL**4**. Each of the four bitlines BL**1**-BL**4** is utilized to read data from the memory cell **104** to one of four separate ports. Additionally, the integrated circuit **100** includes a group **122** of four word lines, WL**1**-WL**4**. Data can be read from the memory cell **104** to one of the ports by selecting one of the bitlines BL**1**-BL**4** and one of the word lines WL**1**-WL**4**. For example, if data is to be read from the memory cell **104** to a first port, then the bitline BL**1** and the word line WL**1** will be selected.
(11) In some embodiments, the integrated circuit includes a respective LIO circuit **106** for each bitline. In the example of FIG. **1**, there are four bitlines. Accordingly there are four LIO circuits **106**. Each of the LIO circuits **106** are identical to each other. Each LIO circuit **106** includes a first sense stage **110**. The first sense stage **110** performs a first sense operation of the corresponding bitline when that bitline is selected for a read operation of the memory cell **104**.
(12) Each of the LIO circuits **106** is coupled to a respective GIO circuit **108**. When a read operation is performed for one of the bitlines, the corresponding LIO circuit **106** senses the data on the bitline and passes the sensed data value, or its logical compliment, to the respective GIO circuit **108**. The GIO circuit **108** then senses the data value provided from the LIO circuit **106** and outputs the data value to the corresponding output port OUT**1**-OUT**4**.
(13) Each GIO circuit **108** includes a second sense stage **114** and a latch **116**. The second sense stage **114** senses the data value received from the LIO circuit **106**. The second sense stage **114** provides the data value to the latch **116**. The latch **116** latches the output OUT of the GIO circuit **108** at the data value sensed by the second sense stage **114**. In this manner, a GIO circuit **108** receives a data value from an LIO circuit **106** and outputs the data value. Receiving the data value from the LIO circuit **106** can include receiving a logical complement of the data value.
(14) It is possible that the integrated circuit **100** can include a large number of LIO circuits **106**. In the case of multiport memory cells, there may be multiple LIO circuits **106** for each column of memory cells in the array of memory cells **102**. If there are four ports per memory cell, then there may be four LIO circuits **106** for each column of memory cells in the memory array **102**. Furthermore, in some cases a memory array **102** may include multiple banks of memory cells in order to reduce parasitic capacitance between bitlines, as will be explained in more detail below in relation to FIG. **3**. The bank configuration breaks each column of memory cells into two or more groups of bitlines. A first half of a column may have a first set of bitlines. The second half of a column may have a second set of bitlines. The integrated circuit may include LIO circuits **106** for each bitline of both sets. Accordingly, there may be a large number of LIO circuits **106** for memory array banking configurations.
(15) One possible solution for reading data from memory cells in multiport or banked configurations, is to include, in each LIO circuit, a complex logic circuit that senses the data from the respective bitline. However, this solution results in a large area usage of the integrated circuit. This is because each LIO circuit would include the complex logic circuit to sense the data from the corresponding bitline. If there is a large number of ports, banks, or both, then the area collectively consumed by the LIO circuits **106** can become quite large.
(16) Embodiments of the present disclosure overcome the area penalty drawbacks of other solutions by providing sensing stages in both the LIO circuits **106** and the GIO circuits **108**. In particular, the LIO circuits **106** each include a first sense stage **110** that is very simple and consumes a very small amount of area. The GIO circuit **108** includes a more complex second sense stage **114**.

(17) The LIO circuits **106** and GIO circuits **108** can also be utilized in banked memory array configurations, as will be described in more detail in relation to FIG. **3**. Accordingly, an integrated circuit **100** including the LIO circuits **106** and the GIO circuit **108** as described in relation to FIG. **1** and subsequent figures, provides reliable multiport or banked memory sensing while consuming less area compared to other solutions.

(18) FIG. **2** is a block diagram of a portion of a memory array **102** of an integrated circuit **200**, according to one embodiment. In particular, FIG. **2** illustrates a single column of memory cells. The column of memory cells includes four memory cells **104***a*-**104***d*. In practice, there may be many more memory cells than four in each column in memory cells. The memory array **102** includes a group of bitlines **120**. In this case, the memory array **102** is a multiport memory array such that there are multiple bitlines for each column of memory cells. The memory array **102** includes, for each row of memory cells, a respective group of word lines **122***a*-**122***d*.

(19) The integrated circuit **200** includes, for each bitline, a respective LIO circuit **106** as described in relation to FIG. **1**. The LIO circuits **106** of FIG. **2** can be substantially identical to those shown in FIG. **1**, including first sense stages **110**. The integrated circuit **200** also includes a plurality of GIO circuits **108**. The GIO circuits **108** of FIG. **2** can be substantially the same as the GIO circuits **108** of FIG. **1**, including the second sense stage **114**. The bitlines **120**, the word lines **122***a*-**122***d*, the LIO circuits **106**, and the GIO circuits **108** cooperate to perform read operations from the memory cells **104***a*-**104***d*.

(20) FIG. **3** is a block diagram of a portion of a memory array **102** of an integrated circuit **300**, according to one embodiment. In FIG. **3**, the memory array is in a banked configuration. In the banked configuration, the memory array **102** is divided into two banks or sub arrays of memory cells.

(21) As mentioned previously, when each column of memory cells includes a large number of memory cells, the bitline or bitlines (in a multiport configuration) for a column of memory cells may become very long. The length of the bitlines can result in a very large parasitic capacitance between the bitlines. In one example, an array of memory cells can include 512 rows of memory cells. Accordingly, each column of memory cells would include 512 memory cells. To reduce the length of the bitlines, the memory array **102** is divided into two or more banks. In the bank configuration, half of the memory cells of each column are in the first bank **103***a* and the other half of the memory cells of each column are in the second bank **103***b*. The result is that the lengths of the bitlines is halved compared to a non-banked configuration. This greatly reduces the parasitic capacitance between the bitlines. Furthermore, each bank **103***a*, **103***b* may include a large number of columns of memory cells. There may be more than two banks of memory cells without departing from the scope of the present disclosure.

(22) In the example of FIG. **3**, a first group **120***a* bitlines is coupled to the memory cells **104***a* and **104***b* of the first bank of memory cells **103***a*. A second group **120***b* of bitlines is coupled to the memory cells **104***c* and **104***d* of the second bank of memory cells **103***b*. In practice, there may be many more than two memory cells in each column in each bank of memory cells. For example, there may be 256 or more memory cells in each column in each bank of memory cells.

(23) The first group **120***a* of bitlines includes a bitline **121***a*, a bitline **123***a*, a bitline **125***a*, and a bitline **127***a*. The second group **120***b* of bitlines includes a bitline **121***b*, a bitline **123***b*, a bitline **125***b*, and a bitline **127***b*. The banked memory array results in a single bitline being separated into two bitlines. For example, bitlines **121***a* and **121***b* would be a continuous single bitline if the memory array was not banked. Because the memory array is banked, the bitlines can be considered as segmented bitlines. Bitlines **121***a* and **121***b* are a singled segmented bitline. Bitlines **123***a* and **123***b* are a singled segmented bitline. Bitlines **125***a* and **125***b* are a singled segmented bitline. Bitlines **127***a* and **127***b* are a singled segmented bitline. Each of the bitlines from the group **120***a* is coupled to a respective LIO circuit **106**. Each of the bitlines from the group **120***b* is coupled to a respective LIO circuit **106**.

(24) There is a single GIO circuit **108** associate with each segmented bitline. Accordingly, the LIO circuits coupled to the bitlines **121***a* and **121***b* are coupled to a single GIO circuit **108**. The two LIO circuits **108** coupled to the bitlines **123***a* and **123***b* are coupled to a single GIO circuit **108**. The LIO circuits **106** coupled to the bitlines **125***a* and **125***b* are coupled to a single GIO circuit **108**. The LIO circuits **106** coupled to the bitlines **127***a* and **127***b* are coupled to a single GIO circuit **108**.
(25) As can be seen in the example of FIG. **3**, the number of LIO circuits **106** can become quite large for banked memory arrays of multiport memory cells. Additionally there may be a large number of columns of memory cells in each bank of memory cells, further increasing the number of LIO circuits **106**. Accordingly, the LIO circuits **106** and the GIO circuit **108** as described in relation to FIG. **1** provide large area savings by moving more complex sense circuitry from the LIO circuits **106** to the GIO circuits **108** because there are fewer GIO circuit **108** than LIO circuits **106**. If the memory array includes four banks instead of two, each segmented bitline will include 4 bitlines and each GIO circuit **108** will be coupled to four LIO circuits **106**. Because increasing the number of memory banks does not increase the number of GIO circuits **108**, the area savings resulting from moving complex circuitry from the LIO circuits **106** to the GIO circuits **108** increases with the increasing number of memory banks.
(26) FIG. **4** is a schematic diagram of an integrated circuit **400** including a memory array **102**, according to one embodiment. FIG. **4** illustrates only a single memory cell **104** of the memory array **102**. In the example of FIG. **4**, the memory cell **104** is an SRAM memory cell including two cross coupled inverters **124***a*, **124***b*, though other types of memory cells can be utilized without departing from the scope of the present disclosure. For simplification only 1 Read port and latch element of a t 6T SRAM cell is shown (pass transistors of 6T element are not shown). Similarly only one LIO **106** and GIO **108** circuit associated to only 1 port is shown in FIG. **4**.
(27) A single bitline BL is coupled to the memory cell **104** via the NMOS transistors N**1** and N**2**. In the example of FIG. **4**, the memory array **102** may be a banked memory array in which the memory array **102** is divided into multiple banks. Each column of memory cells in each bank may be coupled to a single bitline, or multiple bitlines through a multiplexer. Each bitline is coupled to an LIO circuit **106**. A single word line WL is coupled to the gate of the transistor N**2** and can drive multiple columns based on the size of the memory array. The word line WL is connected to a word line driver **134**. And for simplification only one bitcell of a column is shown connect to single bitline.
(28) The bitline BL is coupled to the drain terminal of the PMOS transistor P**1** and to the drain terminal of the NMOS transistor N**3**. The gate of the PMOS transistor P**1** receives a bitline precharge signal BPC. The gate of the NMOS transistor N**3** receives a bitline keep signal BK. The source of the PMOS transistor P**1** is coupled to the high supply voltage VDD. The source of the NMOS transistor N**3** is coupled to ground.
(29) The LIO circuit **106** includes a first sense stage **110**. The first sense stage **110** includes an inverter **126** and an arrangement of NMOS transistors N**4**, N**5** and PMOS P**2** and P**3**. The input of the inverter **126** is coupled to the bitline BL. The output of the inverter **126** is supplied to the gate terminals of the transistors N**5** and P**2**. The gate of the transistor N**4** receives a selection signal SEL. The gate of the transistor P**3** receives the selection signal SELB, which is the logical complement of SEL. The source terminal of the transistor P**3** is coupled to VDD. The source terminal of the transistor N**4** is coupled to ground.
(30) The LIO circuit **106** includes a precharge detector **128** coupled to a dummy bitline DBL and dummy inverter **132**. Dummy circuit elements can be shared across various bitlines of a same port. The LIO circuit **106** also includes a select circuit **130** that generates the selection signals SEL and SELB. The output of the LIO circuit **106** is the drain terminals of the transistors N**5** and P**2**. The output of the LIO circuit **106** is coupled to the GIO circuit **108**.
(31) The GIO circuit **108** (shown for 1 port and 1 global bitline) includes a global bitline GBL and a second sense stage **114**. The second sense stage **114** includes an inverter **140**, a NOR gate **138**,

and a NAND gate **136**. The second sense stage **114** also includes NMOS transistor N**6** and PMOS transistor P**5**. The global bitline GBL is coupled to the output of the LIO circuit **106** into the input of the inverter **140**. The global bitline GBL is also coupled to the keeper circuit **142** and to the drain terminal of the PMOS transistor P**4**. The gate terminal of the transistor P**4** receives a global precharge signal GPC. The source of the transistor P**4** is coupled to VDD.
(32) The output of the inverter **140** is coupled to an input of the NOR gate **138** and the NAND gate **136**. A second input of the NAND gate **136** receives an evaluation signal EVAL. A second input of the NOR gate **138** receives an evaluation signal EVALB, which is the logical complement of EVAL. The output of the NAND gate **136** is coupled to the gate terminal of the PMOS transistor P**5**. The output of the NOR gate **138** is coupled to the gate of the transistor N**6**. The source of the transistor N**6** is coupled to ground. The source of the transistor P**5** is coupled to VDD. The drain terminals of the transistors N**6** and P**5** are coupled together and correspond to the output terminal OUT of the GIO circuit **108**. The GIO circuit **108** also includes a latch **116** that latches the output terminal of the GIO circuit **108** at the most recent value supplied at the drain terminals of the transistors N**6** and P**5**.
(33) The latch **116** is coupled to the keeper circuit **142**.
(34) FIG. **5** is a timing diagram **500** illustrating the timing of various signals generated during a read operation of the memory cell **104** of FIG. **4**, according to one embodiment. The operation of the circuitry of FIG. **4** will be described jointly with the description of FIG. **5**.
(35) FIG. **5** illustrates a clock signal CK. At time t**1**, the clock signal CK transitions from a low logic level to a high lodging value. The rising edge of the clock signal CK causes the bitline keep signal BK, the bitline precharge signal BPC, and the global bitline precharge signal GPC to transition from a high logic level to a low logic level (e.g. ground) at time t**2** in preparation for a read operation of the memory cell **104**. As used herein, the high logic level may correspond to the supply voltage VDD. The low logic level may correspond to ground.
(36) When BK goes low, the transistor N**3** is turned off, thereby decoupling the bitline BL from ground. When BPC goes low, the transistor P**1** is turned on, thereby coupling the bitline BL to VDD. This precharges the bitline BL to VDD in preparation for the read operation. When GPC goes low, the transistor P**4** is turned on, thereby coupling the global bitline GBL to VDD. This precharges the global bitline GBL to VDD in preparation for a read operation.
(37) At time t**3** the bitline BL is precharged to VDD because the bitline precharge signal BPC has caused the bitline to be coupled to VDD. In practice, the bitline BL will begin to increase toward VDD as soon as the transistor P**1** is turned on by BPC. The charging of the bitline BL also causes the charging of the dummy bitline DBL. The charging of the dummy bitline DBL is detected by the detector **128**. The circuit operates under the assumption that if the dummy bitline DBL has reached VDD, then the bitline BL will also have reached VDD. Accordingly, the detector **128** vicariously senses when the bitline BL is precharged by sensing when the dummy bitline DBL is precharged.
(38) At time t**4** BPC and GPC transition from the low logic level to the high logic level responsive to the detector **128** detecting that the dummy bitline is precharged. When BPC and GPC transition to the high logic level, the transistors P**1** and P**4** are turned off. This decouples the bitline BL and global bitline GBL from VDD.
(39) At time t**4**, the word line WL and the selection signal SEL go high responsive to the detector **128** detecting that the dummy bitline DBL is precharged. The word line driver **134** drives the word line WL to VDD. The bank select circuit **130** drives SEL to VDD. When the word line WL goes high, the transistor N**2** is turned on. Depending on the value of the data stored in the memory cell **104**, the bitline BL will either transition to the low logic level or remain at the high logic level. The value of the data stored in the memory cell **104** corresponds to the voltage at the output of the inverter **124***a*. If the data value is 1 (VDD) then the transistor N**1** is conducting and the bitline BL is coupled to ground via N**1** and N**2** (because WL is high). If the data value is 0 (ground), then the transistor N**1** is not conducting and the bitline BL is not coupled to ground. In this case, the bitline

BL remains at the high logic level from the precharge.

(40) In the example of FIG. **5**, the value stored in the memory cell is 1. Accordingly, the bitline BL is coupled to ground and at time t**5** the bitline has transitioned to ground. When the bitline BL goes to ground, the input of the inverter **126** of the LIO circuit **106** goes to ground. The output of the inverter **126** goes high. Because the output of the inverter **126** is high, the transistor P**2** becomes nonconducting and the transistor N**5** becomes conducting. Because the selection signal SEL is high, the transistor N**4** is conducting. Because the complementary selection signal SELB is low the transistor P**3** is conducting. Because P**2** is not conducting, the drains of P**2** and N**5** are decoupled from VDD. Because N**4** and N**5** are conducting, the drains of P**2** and N**5** are coupled to ground. Accordingly, the output of the LIO circuit **106** is ground (low logic level). The operation of the first sense stage **110** corresponds to a first sense operation. The first sense operation senses the data value stored in the memory cell **104** by sensing the complement of the data value stored in the memory cell **104**.

(41) At time t**6**, the global bitline GBL transitions to the low logic level. This is because, as explained above, the output of the LIO circuit **106** has transitioned to the low logic level. It should be noted, that, in practice, the global bitline GBL is coupled to the output of all of the LIO circuits **106** coupled to the various bitlines of the memory array **102**. However, the transistors N**4** and P**3** of the other LIO circuits **106** are not conducting because for them, SEL is low because they are not selected for the read operation. Only one LIO circuit **106** at a time is selected for the read operation. In this case, the LIO circuit **106** shown in FIG. **4** is selected for the read operation. The select circuit **130** of the non-selected LIO circuits **106** output SEL at a low value.

(42) At time t**6**, the evaluation signal EVAL goes high. The evaluation signal EVAL goes high responsive to the word line WL going high but with a delay. The value of the delay is selected to ensure that the global bitline GBL receives the data value (in this case the complement) stored in the memory cell **104**. In practice, the evaluation signal EVAL may go high slightly before or slightly after t**6**.

(43) When GBL goes low at time t**6**, the input of the inverter **140** receives the low logic level. Accordingly, the output of the inverter **140** provides the high logic level. The NAND gate **136** receives the high EVAL signal and the high logic level from the inverter **140**. Accordingly, the NAND gate **136** outputs the low logic level. The NOR gate **138** receives the low EVALB signal and the high logic level from the inverter **140**. Accordingly, the NOR gate **138** outputs a low logic level. The low logic level output by the NOR gate **138** turns off the transistor N**6**, thereby decoupling the drain terminals of the transistors N**6** and P**5** from ground. The low-voltage output by the NAND gate **136** turns on the transistor P**5**, thereby coupling the drain terminals of the transistors N**6** and P**5** to VDD. Because the output OUT of the GIO circuit **108** is coupled to the drain terminals of the transistors N**6** and P**5**, the output of the GIO circuit **108** is the high logic level at time t**7**. This is the value stored in the memory cell **104**. The latch **116** latches the output OUT of the GIO circuit **108** to this value. At t**8**, the evaluation signal EVAL transitions from the high logic level to the low logic level (managed by an internal delay). However, because the latch **116** has latched the output of the GIO circuit **108** at the most recent value, the output OUT of the GIO circuit **108** remains at the high logic level.

(44) At time t**9** the bitline keeper signal BK transitions to the high logic level, timed internally. Similarly, the word line driver **134** drives the word line WL to the low logic level and also the bank select circuit **130** drives the selection signal SEL to the low logic level. The bitline BL is coupled to ground because the transistor N**3** is now turned on. Accordingly, the bitline BL is brought to ground after every read operation. To the contrary, the global bitline GBL is not brought to ground after every read operation. Instead, the global bitline GBL is kept by the keeper circuit **142** at the most recent value between read cycles.

(45) FIG. **6** is a schematic diagram of an integrated circuit **600** including a multiport memory cell **104**. The multiport memory cell **104** is an SRAM memory cell including cross coupled inverters

124*a*, **124***b*. Four bitlines BL**1**-BL**4** and four word lines WL**1**-WL**4** are coupled to the memory cell **104**.
(46) The inverter **124***a* includes the PMOS transistor P**6** and the NMOS transistor N**7** having gate terminals and drain terminals coupled together. The inverter **124***b* includes the PMOS transistor P**7** and the NMOS transistor N**8** having gate and drain terminals coupled together. The source terminals of the transistors P**7** and P**6** are coupled to VDD. The source terminals of the transistors N**8** and N**7** are coupled to ground. The drain terminals of the transistors P**6** and N**7** correspond to the output of the memory cell **104** driving the gate terminals of N**11**, N**13**, N**15** and N**17**. Accordingly, the value of the data stored in the memory cell **104** is the logic level at the drain terminals of the transistors P**6** and N**7**.
(47) The NMOS transistor N**9** is coupled between the output of the inverter **124***b* and the false bitline BLF. The NMOS transistor N**10** is coupled between the output of the inverter **124***a* and the true bitline BLT. The gate terminals of the transistors N**9** and N**10** receive a word line write signal WLW for writing data to the memory cell **104**. Accordingly, N**9**, N**10**, BLF and BLT are utilized only in write operations of the memory cell **104**.
(48) The NMOS transistors N**11**, N**13**, N**15**, and N**17** have source terminals coupled to ground and gate terminals coupled to the output of the memory cell **104**. The NMOS N**12**, N**14**, N**16**, and N**18** are coupled between a respective bitline BL**1**-BL**4** and a respective one of the transistors N**11**, N**13**, N**15**, and N**17**. The gate terminals of N**12**, N**14**, N**16**, and N**18** are coupled to a respective word line WL**1**-WL**4**. The pair of transistors coupled to the bitlines BL**1**-BL**4** perform the same function as the transistors N**1** and N**2** described in relation to FIGS. **4** and **5**.
(49) Each bitline BL**1**-BL**4** is coupled to a respective LIO circuit **106***a*-**106***d*. The LIO circuits **106***a*-**106***d* are identical to each other. Only the LIO circuit **106***d* coupled to the bitline BL**4** is shown in detail in FIG. **6**. The LIO circuits **106***a*-**106***d* are identical to the LIO circuit **106** of FIG. **4**. Each of LIO circuits **106***a*-**106***d* is connected to a global bitline of a respective GIO circuit **108**. The LIO circuits **106***a*-**106***d* and the GIO circuits **108***a*-**108***d* of FIG. **6** function substantially identical to the LIO circuit **106** and GIO circuit **108** shown and described in relation to FIGS. **4** and **5**.
(50) When data is to be read from the memory cell **104** to one of the ports, the corresponding wordline and bitline are selected. The selected wordline driver drives the wordline high. The selected bitline is precharged in preparation for the read operation, as described in relation to FIGS. **4** and **5**. The corresponding LIO **106** receives the high SEL signal and the data from the memory cell is sensed by the first sense stage **110** and passed to the global bitline GBL of the GIO circuit **108**, as described in relation to FIGS. **4** and **5**. The GIO circuit **108** senses the data value on the global bitline GBL and outputs the data value at the output OUT, as described in relation to FIGS. **4** and **5**.
(51) FIG. **7** is a flowchart of a method **700** for reading data from a memory cell, according to one embodiment. At **702**, the method **700** includes storing a data value in a memory cell, according to one embodiment. At **704**, the method **700** includes selecting the memory cell for a read operation, according to one embodiment. At **706**, the method **700** includes outputting the data value to a bitline coupled to the memory cell, according to one embodiment. At **708**, the method **700** includes sensing the data value from the bitline with a local I/O circuit coupled to the read that line, according to one embodiment. At **710**, the method **700** includes providing the data value from the local I/O circuit to a global I/O circuit by selectively enabling the local I/O circuit, according to one embodiment. At **712**, the method **700** includes sensing the data value with the global I/O circuit, according to one embodiment. At **714**, the method **700** includes outputting the data value from the global I/O circuit, according to one embodiment.
(52) The various embodiments described above can be combined to provide further embodiments. These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the

claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. An integrated circuit, comprising: a memory cell configured to store a data value; a plurality of bitlines each coupled to the memory cell; a plurality of local I/O circuits each coupled to the memory cell by a respective bitline and including a first sensing stage configured to sense the data value from the bitline when the respective bitline is selected for a read operation; and a plurality of global I/O circuits each coupled to a respective local I/O circuit and each including: a respective global bitline that can be selectively coupled to receive the data value from the local I/O circuit coupled to the bitline selected for the read operation; and a respective second sensing stage including an evaluation circuit configured to sense the data value output by the respective global bitline, wherein the first sensing stage of each of the local I/O circuits includes a first inverter having an input coupled to the respective bitline, wherein the respective second sensing stage includes a second inverter having an input coupled to the respective global bitline, and wherein the first inverter is different from the second inverter.
2. The integrated circuit of claim 1, wherein each respective second sensing stage includes: a first logic gate having: a first input coupled to an output of the second inverter; a second input configured to receive a first evaluation enable signal; and an output coupled to an output of a respective one of the global I/O circuits.
3. The integrated circuit of claim 2, wherein each respective second sensing stage includes: a second logic gate having: a first input coupled to the output of the second inverter; a second input configured to receive a second evaluation enable signal; and an output coupled to the output of the respective one of the global I/O circuits.
4. The integrated circuit of claim 3, wherein the global I/O circuit includes a latch coupled to the outputs of the first and second logic gates.
5. The integrated circuit of claim 4, wherein the first logic gate is a NAND gate and the second logic gate is a NOR gate.
6. The integrated circuit of claim 1, wherein the first sensing stage includes a selection circuit coupled to an output of the first inverter and configured to selectively provide the data value to the respective global bitline responsive to a selection signal by inverting a signal output by the first inverter.
7. The integrated circuit of claim 6, wherein the selection circuit includes: a first NMOS transistor having a gate terminal coupled to the output of the first inverter and a drain terminal coupled to the respective global bitline; and a first PMOS transistor having a gate terminal coupled to the output of the first inverter and drain terminal coupled to the drain terminal of the first NMOS transistor.
8. The integrated circuit of claim 7, wherein the selection circuit includes: a second NMOS transistor having a drain terminal coupled to a source terminal of the first NMOS transistor and a source terminal coupled to ground; and a second PMOS transistor having a drain terminal coupled to a source terminal of the first PMOS transistor and a source terminal coupled to a high voltage supply.
9. An integrated circuit, comprising: a memory array including a column separated into a first bank of memory cells and a second bank of memory cells; a segmented bitline associated with the column of memory cells, the segmented bitline including a first bitline segment coupled to the first bank of memory cells and a second bitline segment coupled to the second bank of memory cells, the first bitline segment and the second bitline segment being discontinuous from each other; a first local I/O circuit coupled to the first bitline segment and including a first sensing stage configured to sense a data value from a first memory cell of the first bank when the first memory cell is selected

for a read operation; a second local I/O circuit coupled to the second bitline segment and including a first sensing stage configured to sense a data value from a second memory cell of the second bank when the first memory cell is selected for the read operation; and a global I/O circuit coupled between the first local I/O circuit and the second local I/O circuit and including: a global bitline that can be selectively coupled to receive the data value from the first local I/O circuit or the second local I/O circuit selected for the read operation; and a second sensing stage including an evaluation circuit configured to sense the data value output by the global bitline.

10. The integrated circuit of claim 9, wherein the memory array is an SRAM memory array.

11. The integrated circuit of claim 9, wherein the memory cells are multi-port memory cells.

12. The integrated circuit of claim 11, wherein the first sensing stage is configured to sense the data value from a selected memory cell by sensing the data value from a corresponding bitline.

13. The integrated circuit of claim 12, wherein the first sensing stage includes a first inverter having an input coupled to the bitline, wherein the second sensing stage includes a second inverter having an input coupled to the global bitline.

14. A method, comprising: storing a data value in a memory cell coupled to a plurality of bitlines each coupled to a respective local I/O circuit, each local I/O circuit coupled to a respective global I/O circuit; selecting the memory cell for a read operation; outputting the data value to a selected bitline of the plurality of bitlines coupled to the memory cell; sensing the data value from the selected bitline with the local I/O circuit coupled to the selected bitline, the local I/O circuit coupled to the selected bitline corresponding to a selected local I/O circuit, wherein each global I/O circuit includes: a respective global bitline that can be selectively coupled to receive the data value from the respective local I/O circuit coupled to the bitline; and a respective second sensing stage including an evaluation circuit configured to sense the data value output by the respective global bitline, the global I/O circuit coupled to the selected local I/O circuit corresponding to a selected global I/O circuit, a first sensing stage of each of the local I/O circuits including a first inverter having an input coupled to the respective bitline, the respective second sensing stage including a second inverter having an input coupled to the respective global bitline, the first inverter being different from the second inverter; providing the data value from the selected local I/O circuit to the global I/O circuit coupled to the selected local I/O circuit by selectively enabling the selected local I/O circuit; sensing the data value with the selected global I/O circuit; and outputting the data value from the selected global I/O circuit.

15. The method of claim 14, wherein sensing the data value with the selected local I/O circuit includes inverting the data value with a first inverter of the selected local I/O circuit.

16. The method of claim 15, wherein providing the data value from the selected local I/O circuit to the selected global I/O circuit includes providing the data value from the selected local I/O circuit to a global bitline of the selected global I/O circuit.

17. The method of claim 16, wherein sensing the data value with the selected global I/O circuit includes inverting the data value from the global bitline.

18. The method of claim 14, further comprising: discharging the selected bitline between every read operation by coupling the selected bitline to ground between every read operation; and keeping, between every read operation, the global bitline at a voltage corresponding to the data value of a most recent read operation.